



Final Bachelor's Thesis
Bachelor's in Economics and Management

The Limits of Wealth: Exploring the Impacts of GDP and Other Socioeconomic Factors on National Happiness Levels - an Empirical Analysis

By: SEMMAR Dhikra

Supervisors:
JOUF Chouaib

24/05/2024

Introduction:

A recurring question among people, be they economists, construction workers, or quantum physicists, is whether money can buy happiness—an age-old question that has been studied countless times. So, does it really?

In this study, we will dive deeper into the core of this question and explore the subjective and varied nature of its answers across nations by examining the factors that influence people's responses, specifically the socioeconomic factors.

Our dataset consists of the Happiness Score for 153 countries along with the factors used to explain the score. The Happiness Score is a national average of responses to the main life evaluation question asked in the Gallup World Poll (GWP), which employs the Cantril Ladder. The Happiness Score is explained by the following factors: GDP per capita, Healthy Life Expectancy, Social Support, Freedom to Make Life Choices, Generosity, Corruption Perception, and Residual Error.

Through this study, we will attempt to answer the question, "Does money buy happiness?" in two parts:

Part 1: Does an increase in GDP per capita cause an increase in happiness?

We will explore this question under two hypotheses using linear regression, tests of correlation, and an analysis of the effect of this variable on the overall happiness score. After studying the effects of GDP, we will examine the impacts of each socioeconomic factor to determine which are most significant for the happiness score across all countries in our dataset.

Part 2: Is there a limit to the contribution of wealth to happiness?

In this part, we will investigate whether there is a point at which an increase in personal wealth diminishes the happiness score, by studying the direction of marginal utility. We will explore if there is a threshold beyond which an increase in wealth no longer contributes to happiness.

Additionally, we will seek to understand these results and pinpoint the reasons behind them through specific case studies based on geopolitical context.

Using our dataset, along with economic literature and past research similar to this one, we aim to understand the impact of socioeconomic factors on our perception of happiness. This understanding will enable us to guide public policies that encourage prosperity driven by financial means and make progress on a global level in terms of welfare.

The purpose of our study is to identify the factors that significantly impact the happiness score of a given country, whether positively or negatively. By understanding these factors, we aim to manipulate the outcome through informed public policies and strategic reorganization of institutions and infrastructures to enhance overall happiness.

Our methodology involves analyzing the most impactful variables influencing the happiness score and determining if increasing these variables benefits the happiness index. Based on these findings, we will identify suitable public policies and laws to enhance happiness. In addition, we will compare the top 10 countries with the highest happiness scores to the 10 countries with the lowest happiness scores, pinpointing major differences and formulating strategies to bring the lowest scoring countries closer to the highest by emulating successful policies and practices. By following this methodology, we aim to provide actionable insights for policymakers to enhance the happiness of their populations through targeted interventions and improvements in key areas.

Context:

Before we delve into the study, it is mandatory for us to understand what each variable used in the dataset means, and contains.

Happiness Score: The Gallup World Poll (GWP) measures subjective well-being (SWB) through a national average response to a simple question: "On a scale from 0 to 10, where 0 is the worst possible life and 10 is the best, where do you personally stand in life right now?" This measure, also known as the Cantril Life Ladder or Life Ladder, spans from 2005 to 2019 and offers insight into individuals' overall life satisfaction and well-being.

GDP per capita: The measure of a country's economic output per person, adjusted for purchasing power parity (PPP) at constant 2011 international dollar prices. It represents the average income per person in a country.

Healthy Life Expectancy (HLE): The expected number of years of life in good health at birth. It reflects the average number of years a person can expect to live in good health, taking into account mortality and morbidity rates.

Social support: The perceived availability of relatives or friends to count on in times of trouble. It is based on responses to the question, "If you were in trouble, do you have relatives or friends you can count on to help you whenever you need them, or not?"

Freedom to make life choices: The perceived satisfaction with the freedom to choose what to do with one's life. It is based on responses to the question, "Are you satisfied or dissatisfied with your freedom to choose what you do with your life?"

Generosity: The residual of regressing the national average of response to the question, "Have you donated money to a charity in the past month?" on GDP per capita. It represents the extent of charitable giving relative to a country's economic prosperity.

Corruption Perception: The perceived level of corruption in government and business sectors. It is the national average of responses to questions regarding the prevalence of corruption in government and business.

Residual error: The discrepancy between observed and predicted values in a statistical model. It represents unexplained variability in the dependent variable after accounting for the effects of independent variables.

Hypotheses:

“Do personal wealth and Happiness correlate positively?”

H1: There is no significant relationship between GDP per capita and national happiness levels. This hypothesis could potentially indicate that economic prosperity alone does not guarantee greater happiness.

“Is there a turning point after which a greater income has a negative impact on one’s happiness?”

H2: There is a limit to financial contribution on the overall happiness score. This hypothesis suggests that gdp per capita can have an adverse effect on the happiness score, implying a turning point in the relationship between the variables.

Methodology:

1. Data Exploration and cleaning:

At first glance, the datasets look very extensive in the information and details they offer, which is good to see since an information-dense dataset is always preferred, and gives overall more insight in the results at the end.

However, I was quick to notice a heterogeneity within the: variable names, amount of observations, the variable types (integers, strings...) as well as the order that these variables were organized across the different datasets, the latter describing different years in which these observations were noted.

Consequently, i started to organize, clean, and homogenize each dataset so that my analysis will be simpler to attempt, and to minimize the chances of running into bugs within my code, and that, through the steps mentioned below:

a. Data Loading:

Initially, data from multiple CSV files representing yearly datasets were loaded into the environment, the structure of each dataset (2015...2022) was examined, providing insights into the composition of the dataset, including the number of observations, variables, and their data types. This step helps in understanding the underlying structure of the data.

b. Standardizing Column Names:

Standardizing column names is a crucial step during the cleaning process, to ensure consistency across datasets, especially when merging or analyzing them together (something we will get to later on during the report). For example, "overall.rank" was renamed to "happiness.rank" to maintain consistency.

c. Ranking and Region Column Addition:

Some datasets lacked important columns like "Region", which were present in other datasets. Therefore, these missing columns were merged using existing columns from other, more complete datasets, to ensure completeness.

Rankings based on happiness scores were added to datasets "twenty" and "twentyone" as they were absent initially, allowing us to compare them with other years' datasets.

d. Variable Transformation:

The creation of "Explained By" (EB) variables involved multiplying predictor variables with their corresponding coefficients obtained from linear regression models. This step helps in understanding how much each predictor variable contributes to explaining the variation in the target variable (Happiness score).

In dataset "twentytwo", certain variables were converted from character to numeric format to enable numerical analysis. This conversion ensures that the variables are treated as numerical rather than categorical or character data types.

e. Isolation of Remaining Variables:

Some variables were not uniformly present across all datasets. These variables were isolated and stored separately for each dataset (in case we will need them during our analysis without them getting in the way during the coding process) to ensure that the datasets have a consistent structure, we called these variables "remnants".

This step involves segregating variables like "Standard.Error", "Dystopia.Residual", "Lower.Confidence.Interval", etc., which were not uniformly present across all datasets.

f. Final Column Names Verification and ordering:

We verified the consistency of column names across all datasets to ensure that subsequent analyses are conducted smoothly, by using a function to identify set differences between variable names across datasets. Any discrepancies in column names were identified and resolved to maintain consistency.

After which, we ordered the variables in each dataset using the same order as the dataset 2015 for simplicity, which will allow us to make the right comparisons during our analysis.

g. Merging datasets into a single dataframe:

In this step, I decided it would be optimal for the analysis to combine all of the different datasets into a single comprehensive dataset called "masterdata". afterwards, we modified two variables, "GDP.pc" and "Healthy.life.expectancy", in the Master dataframe. This modification is based on a condition where if the "Year" column value equals either 2020 or 2021, the values of "GDP.pc" and "Healthy.life.expectancy" are multiplied by 0.1, because of an anomaly in which the

observations were formatted, we decided to scale them down so they match the other formats of this variable in each year preceding. Following the application of these transformations, the adjusted Master data frame is saved to a new CSV file named "master.o.csv" using the write.csv() function.

2. Data analysis:

To test H1 (concerning the socioeconomic impact on happiness which we predict to be overall positive, aside from a few exceptions) we started by running a correlation between happiness score and all our independent variables, to measure the strength and direction of the relationship between each two variables in a general manner. Next we ran a linear regression to model the relationship between our dependent variable, or the explained one (happiness score) and independent variables to find the best-fitting linear equation that predicts the dependent variable based on the independent variables regrouping all socioeconomic factors in our dataset.

We enhanced the performance of our regression model by handling heteroskedasticity (Appendix 1) and testing for normality of the distribution of residuals and multicollinearity. Heteroskedasticity was successfully handled using squared residuals and weighting the model, by using WLS (Weighted least squares model) instead of OLS, this way we can account for heteroscedasticity by assigning different weights to different observations. This can be observed through the Breusch-Pagan Test results.

We eventually spotted a lack of normality (Appendix 2), which we initially expected to be a product of multicollinearity, due to their frequent coexistence in linear regressions. However, after using the VIF method, which consists of measuring how much the variance of a regression coefficient is inflated due to multicollinearity with other predictors, we did not detect any multicollinearity within our model. Consequently, the absence of normality can be disregarded due to the relatively high number of observations in our database, therefore, it is possible for us to say that our model is asymptotically normal, thanks to the Shapiro Test results.

As an additional step in our analysis, and given our previous tests suggesting a potential absence of normality, we decided to assess the reliability of our model through Ridge regression and an RMSE test. Firstly, the Ridge regression, a type of linear regression that includes a regularization term to handle multicollinearity by shrinking the regression coefficients, enhances the stability and interpretability of the model. Additionally, The Root Mean Square Error (RMSE) test, which measures the differences between predicted and actual values, provides an indicator of model accuracy, with lower values indicating a better fit.

Our results yielded an absence of multicollinearity and proved once and for all the extensive reliability of our model.

To test our second hypothesis; H2 (concerning the existence of a limit to contribution of wealth and possibly other factors to happiness in the world) our approach was to search for a breaking point that questions the linearity of the relation between a given factor and happiness scores, which takes into account absolute change in direction (from upwards to downwards for instance) and stagnation (sudden horizontal setting). To do so, we started by

segmented regressions (Appendix 4-5) for each independent variable. *Segmented regressions*, also known as piecewise or breakpoint regressions, involve dividing the data into distinct segments and fitting separate linear regressions to each segment. This technique is useful for identifying and modeling changes in relationships at specific points in the data. Then, to best challenge the breaking points we ran a GAM regression (Appendix 6), which is a flexible approach that allows for non-linear relationships between the predictors and the response variable. It extends linear models by allowing non-linear functions, typically smooth splines, for each predictor, enabling the modeling of complex patterns in the data while maintaining interpretability.

Results:

Part 1:

1. Correlation results

	Happiness	GDP	Social	Healthy	Freedom	Perceptions	Generosity
Happiness.score	1.00000000	0.69228141	0.575861830	0.69516634	0.415888076	0.08838950	0.04817092
GDP.pc	0.69228141	1.00000000	0.470250976	0.69600400	0.236776934	0.03696084	-0.04797382
Social.support	0.57586183	0.47025098	1.00000000	0.52091787	-0.007037024	-0.27475132	0.19807481
Healthy.life.expectancy	0.69516634	0.69600400	0.520917865	1.00000000	0.283121207	0.13668427	-0.04968633
Freedom	0.41588808	0.23677693	-0.007037024	0.28312121	1.00000000	0.70969300	-0.32042953
Perceptions.of.corruption	0.08838950	0.03696084	-0.274751319	0.13668427	0.709692997	1.00000000	-0.51157052
Generosity	0.04817092	-0.04797382	0.198074806	-0.04968633	-0.320429534	-0.51157052	1.00000000

According to our correlation results (Appendix 3), we can notice a few interesting dynamics between variables. For instance, the analysis reveals that the Happiness Score has a positive and quite high correlation with Social Support, Freedom, healthy life expectancy and GDP per capita , indicating that increases in these variables tend to correspond with higher happiness scores.

Studies like [*“Multiple Measures of Family and Social Support as Predictors of Psychological Well-Being: An Additive Approach”* by Hema O. Mason, 2016](#), have examined the role income and social support--in predicting change in happiness among adults across a 10-year period. According to this study, Income had a small, positive impact on happiness. In contrast, family social support, measured by 3 subscales, Cohesion, Expressiveness, and Conflict, showed a substantial, positive association with concurrent happiness, even after controlling for income.

Additionally, [*Chloe R. Grabanski, Adedoyin Okanlawon, and Raina V. Lamade, found in their 2022 study “Psychosocial Factors Associated with Happiness”*](#), a positive relationship between freedom and happiness by examining cultural differences. In individualistic Western cultures, where personal freedom and achievements are highly valued, happiness is often linked to high arousal positive emotions like excitement and self-esteem. In contrast, collectivistic cultures prioritize social harmony and experience happiness through low arousal emotions like calmness. Additionally, political freedom and good governance further reinforce the link between freedom and happiness, suggesting that environments promoting personal and political freedoms contribute significantly to overall well-being.

When it comes to healthy life expectancy, we notice a high positive coefficient, in both results regarding the linear regression and correlation matrix.

According to past research on this topic, health usually stems from a high level of happiness, not the opposite. Which leads us to believe that the healthy life expectancy factor is a byproduct of increased happiness, and not the other way around.

For GDP per capita, The study [*"Economic growth and subjective well-being: reassessing the easterlin paradox"*](#) conducted by Betsey Stevenson and Justin Wolfers in 2008, finds a strong positive relationship between high income and happiness. As countries' GDP per capita increases, so does the average subjective well-being of their citizens. This correlation holds true across different countries and within individual nations over time. Wealthier individuals tend to report higher levels of happiness, indicating that absolute income significantly impacts well-being. These findings challenge the *Easterlin Paradox* and suggest that economic growth can effectively enhance national happiness, which is something we'll get to later on in the report.

Furthermore, GDP per capita has a strong positive correlation with Healthy Life Expectancy, this seems quite intuitive to say, as GDP per capita impacts healthy life expectancy by enhancing socioeconomic conditions that improve health outcomes. According to the study [*"Socioeconomic Disparities in Health: Pathways and Policies"*](#) by Nancy E. Adler and Katherine Newman, 2002, higher income/wealth allows for better healthcare access, higher quality medical services, and improved living conditions.

On a bigger scale, wealthier nations can invest in advanced healthcare infrastructure, cleaner environments, and healthier lifestyles. This economic affluence enables people to afford nutritious food, access preventive healthcare, and live in safer neighborhoods, directly correlating with increased healthy life expectancy.

On another note, it is interesting to see the negative correlation between Generosity, and perception of corruption, this could be due to the idea that environments characterized by higher levels of corruption may heighten individualistic tendencies as people might focus more on personal gain and self-protection in response to a lack of trust in societal institutions and governance.

2. Linear Regression results

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)	
(Intercept)	1.82584	0.07854	23.247	< 2e-16	***
GDP.pc	0.88998	0.06471	13.753	< 2e-16	***
Social.support	0.88563	0.07534	11.755	< 2e-16	***
Healthy.life.expectancy	1.38710	0.12821	10.819	< 2e-16	***
Freedom	1.98026	0.12492	15.852	< 2e-16	***
Perceptions.of.corruption	-0.44262	0.10355	-4.275	2.06e-05	***
Generosity	0.63171	0.12983	4.866	1.29e-06	***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

We can notice a few differences in outcomes of the two analyses, while Both correlation and linear regression results suggest that factors such as GDP per capita, Social Support, Healthy

Life Expectancy, and Freedom are positively associated with Happiness Score, there is definitely a variation in the numerical results obtained.

Firstly, we can see that all of our coefficients are very significant, meaning that every one of these coefficients has an impact on our dependent variable, the Happiness score.

It is interesting to see that the effect of GDP per capita has increased to 0.89 points, meaning that one unit's increase in GDP.pc increases happiness by 0.89 units. This is noticeable in the variable of social support as well, jumping from 0.57 in the correlation to 0.86 during the linear regression, giving it a more important impact on happiness. Same for Generosity, which increased from 0.04 points -a negligible value- to 0.63.

Moreover, some variable effects changed drastically from the correlation analysis to the linear regression, this is especially true for the case of the variable "perception of corruption", that went from a positive, yet insignificant value of 0.088, to the negative but much more important value of -0.44 post linear regression, this means that a unit of "perception of corruption" has an opposite impact on happiness, decreasing it by 0.44 points. This relationship is quite logical given societal context.

The results of the linear regression align much better with what was anticipated at the earlier stages of the research, proving therefore that a correlation analysis is a good way to find relations between variables initially, but is insufficient at finding patterns and effects below the superficial level on its own. Therefore we favor linear regressions for their simplicity, straightforwardness and ease of computation especially given a performant and unbiased dataset such as our own.

Finally, we confidently reject our hypothesis, and say that a higher revenue can definitely increase happiness within individuals across countries.

Conclusion : Part 1

According to our computations, we have found an interesting relation between socioeconomic factors, and people's perception of happiness.

Our analysis demonstrates that factors such as Social Support, Freedom, Healthy Life Expectancy, and GDP per capita are positively associated with happiness, while higher Perceptions of Corruption negatively impact it. The regression analysis underscores these findings, showing significant effects of these variables on happiness. Thus, economic prosperity, robust social support systems, and lower corruption levels are critical for enhancing individual well-being.

Part 2:

For the second part of our research, we attempted to find out whether or not that there was a point in which a rise in GDP per capita would have a stable, or adverse effect on the happiness score, in other words, detecting a potential limit in the contribution of GDP per capita on the happiness score.

To showcase this, we used a segmented regression analysis, also known as piecewise regression. It is a statistical technique used to model relationships between variables when

the relationship is not linear across the entire range of the data. Instead of fitting a single regression line to all the data points, segmented regression involves fitting multiple regression lines to different segments, or intervals, of the data.

This method is used to identify braking points, where the relationship between variables changes, be it positively or negatively.

Coefficients of the linear terms:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	3.24208	0.07332	44.217	<2e-16 ***
GDP.pc	2.30919	0.07897	29.242	<2e-16 ***
U1.GDP.pc	-1.65627	0.32859	-5.041	NA

Results showed that our model yielded No breaking or pivotal points, this can be seen in our U1 coefficient (change-in-slope coefficient) concerning the GDP per capita segmented regression, the returned result is an NA (short for not available), this means that the breaking point is undetectable, which means it's either negligible or nonexistent, therefore the trend never really changes direction.

To make sure that our results were correct and trustworthy, We decided to run a different test, this time a GAM regression.

GAM (Generalized Additive Model) regression is a statistical modeling technique that extends traditional linear regression by allowing for non-linear relationships between the predictor variables and the response variable. In GAM regressions, the relationship between the response variable and each predictor variable is represented as a smooth, non-linear function.

So instead of linearity assumption and the context adaptation to it, we chose to allow our generalized additive model to detect the linearity on its own, which it effectively did, our GA-model provided results that showed, indeed, smooth and linear relationships between the dependent and independent variables. and with this result it's safe to say that we can't challenge the linearity of any of the studied relationships / correlations. In simpler words, we can't predict when money (and other socio economic factors) stops increasing happiness levels whatsoever.

This leads us to reject our initial hypothesis, concluding thereby that there is NO limit to revenue contribution on happiness.

Conclusion: Part 2

In the second part of our research, we explored whether there is a point at which rising GDP per capita no longer boosts happiness scores. Using segmented regression analysis, we found no such breaking point; the relationship remained consistently linear, this was confirmed by a GAM regression. Thus, our study concludes that there is no detectable limit to GDP's positive impact on happiness within the data we analyzed.

Discussion:

This research has helped us know more in detail about the contributions and impacts of socioeconomic factors on happiness, but most importantly, about wage contribution to happiness.

This led us to a concept known as the Easterlin Paradox, proposed by Richard Easterlin in the 1970s. this concept touches two angles of economics, Macro and Micro,

Macro-Level (Across Countries and Over Time):

When examining different countries, Easterlin found that richer countries generally reported higher average happiness levels than poorer countries. However, this relationship is less pronounced once a certain threshold of wealth is reached. On the other hand, One of Easterlin's key findings was that over time, significant increases in a country's GDP per capita do not necessarily correspond to increases in average happiness. In other words, as countries become wealthier, their citizens do not consistently report higher levels of happiness.

Micro-Level (Within Countries):

Easterlin observed that within a given country, individuals with higher incomes tend to report higher levels of happiness. This implies a positive correlation between personal income and happiness at a specific point in time. An important factor is relative income—how one's income compares to others. People derive satisfaction not just from their absolute income but also from their income relative to their peers. This social comparison can influence perceived happiness.

Easterlin observed that within a given country, individuals with higher incomes tend to report higher levels of happiness. This implies a positive correlation between personal income and happiness at a specific point in time, which was proven by our calculations throughout the study, and as time goes by and GDP per capita increases, be it through inflation or through economic expansion, it was revealed after testing that the happiness score, while it increases, does so at a slower rate than that of the growth of personal wealth.

All in all, the Easterlin paradox aligns with our findings, from a global macroeconomic point of view, to an individual and personal perception of happiness.

Policy Implications:

One of the biggest inspirations of this topic of research was Professor Francis Munier, a distinguished researcher whose work focuses on the economics of happiness, the economics of innovation and knowledge, and the organization of firms. His in-depth studies in the field of happiness economics highlight the importance of considering individual and collective well-being in the formulation of public policies.

The goal of studying the socioeconomic factors playing a role in happiness, is to be able to understand the dynamics that go into enhancing global welfare across the world, one country at a time.

This can be done through:

- *Incorporating Happiness Metrics in Policy Evaluation:*

As mentioned in the article [“L'économie du bonheur: quelques faits saillants et données empiriques en Europe”](#) by Laetitia Dillenseger and Francis Munier, 2019, policymakers ought to go beyond traditional economic indicators like GDP, and incorporate happiness and well-being metrics into their evaluations this will provide a more realistic understanding of societal progress and help identify areas where economic growth does not necessarily translate into improved quality of life.

- *Income and Wealth Distribution:*

Progressive taxation and social welfare programs can help redistribute wealth more equitably, potentially leading to higher overall happiness levels, particularly for those in lower income brackets.

- *Healthcare and Mental Health Services:*

As shown before in our analysis, health and healthcare are closely linked to happiness, and accessible and high-quality healthcare services are crucial when it comes to upgrading the national levels of happiness through enhancing healthcare infrastructures and supporting healthcare professionals through various means, but also acknowledging the importance of mental health, and giving individuals the psychological help they need.

- *Education and Lifelong Learning:*

Education contributes positively to individual well-being by giving people the space and tools to express their opinions, but also encouraging the personal development necessary to attract economic opportunities.

It is a given and undeniable fact that [unemployment has a negative impact on individuals' happiness](#)¹, and by creating policies that favor education policymakers can support agents to create income and find prosperity in their level of wealth.

- *Social Relationships and Community Building:*

As previously proved during our research, Strong social ties and a sense of community are important factors when it comes to increasing happiness. Therefore, policies that encourage community engagement and social interaction can be beneficial these policies can include investing in community centers, supporting family-friendly policies, but also developing interventions through community-based organizations along with local governments, to promote well-being at the community

level, all while taking into account the [cultural preferences and values of different populations within the community](#)².

- *Environmental Policies:*

Environmental quality definitely impacts well-being, studies such as ["Would you be happier living in a greener urban area? A fixed-effects analysis of panel data"](#), conducted by Mathew P White, Ian Alcock, Benedict W Wheeler and Michael H Depledge, came to supporting results: *"..Controlling for individual and regional covariates, we found that, on average, individuals have both lower mental distress and higher well-being when living in urban areas with more green space."*

Policies aimed at reducing pollution, ensuring access to green spaces in urban planning and promoting environmental conservation have been shown to enhance quality of life, thereby enhancing happiness.

- *Measuring and Monitoring Well-being:*

Researchers such as F.Munier have advocated for the regular collection and analysis of well-being data to inform policy decisions, which can allow economists and policymakers to study more concretely other potential factors that can impact the happiness level given within a country, and keep a close watch on the macroeconomic changes and how these changes translate into the microeconomic, individual sphere.

Limitations:

A few things that didn't allow us from studying the question properly would be issues concerning the dataset. while this dataset is incredibly broad and detailed, it was lacking in a few aspects:

- The time frame within which the observations are provided is quite tight, while an 8 year time period can be very beneficial for studying trends, it isn't as efficient at showcasing the changes in a country's happiness overtime for our time-constrained dataset.
- Due to the big amount of observation (countries), we couldn't pinpoint atypical paths of specific countries to trace them back to economic / political events that have influenced their trends specifically, due to our priority being the classic trend over time to reach more global conclusions. It also would have taken us much longer to backtrack and research what could have potentially been multiple events in a single country, given that our dataset concerns around 150 different countries.
- The dataset contains an era of economic crisis, which was COVID 19. Due to the tight time period, it is harder for us to study the long term effects that the pandemic had on individuals.
- To study the microeconomic aspect of the easterlin paradox, it would've been helpful to have an extra variable representing relative income, or rather *a self comparison factor*
- When studying subjective and psychological responses such as happiness, or comfort, it is difficult to acquire unbiased and constant results, humans go through mood variations, and are subjects to the environment around them, which renders

subjective answers such as the one studied in our model hard to quantify. However, this doesn't necessarily have to be a limitation, it can be interesting to revisit this aspect of the study through a behavioral lens, and incorporate nudges or study the cognitive biases and heuristics that individuals are subjected to during their valuation in the data collection process.

Summary:

Our study analyzed the impact of socioeconomic factors on happiness using correlations and linear regressions, accounting for heteroskedasticity and testing for multicollinearity. We found significant positive relationships between happiness and factors like GDP per capita, Social Support, Healthy Life Expectancy, and Freedom, with perceptions of corruption negatively impacting happiness.

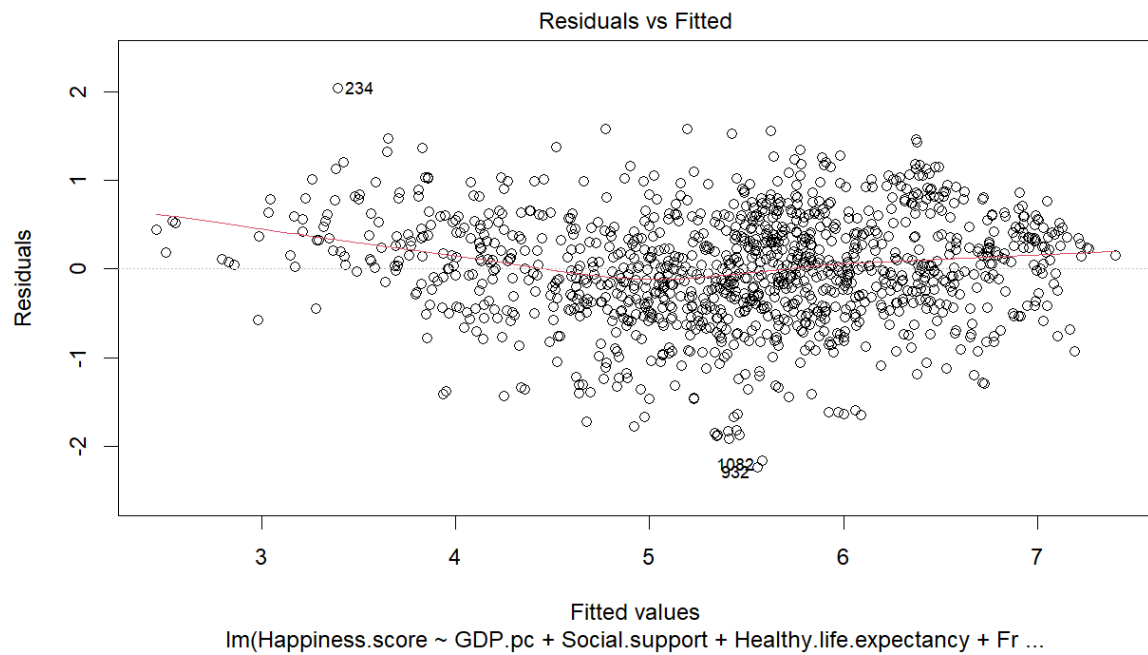
Further analysis using segmented and Generalized Additive Models (GAM) regressions indicated no breaking points, suggesting a consistently positive relationship between GDP per capita and happiness. These findings highlight the importance of integrating happiness metrics with traditional economic indicators, promoting equitable wealth distribution, investing in healthcare and education, supporting community building, and implementing environmental policies to enhance overall well-being.

As our journey through data manipulation and analysis draws to a close, we find ourselves more understanding of the undeniable truth that socioeconomic factors profoundly dictate the aspects of human happiness, where the prosperity and comfort brought by both material and psychological circumstances shape people's perception of what it's like to be satisfied from life.

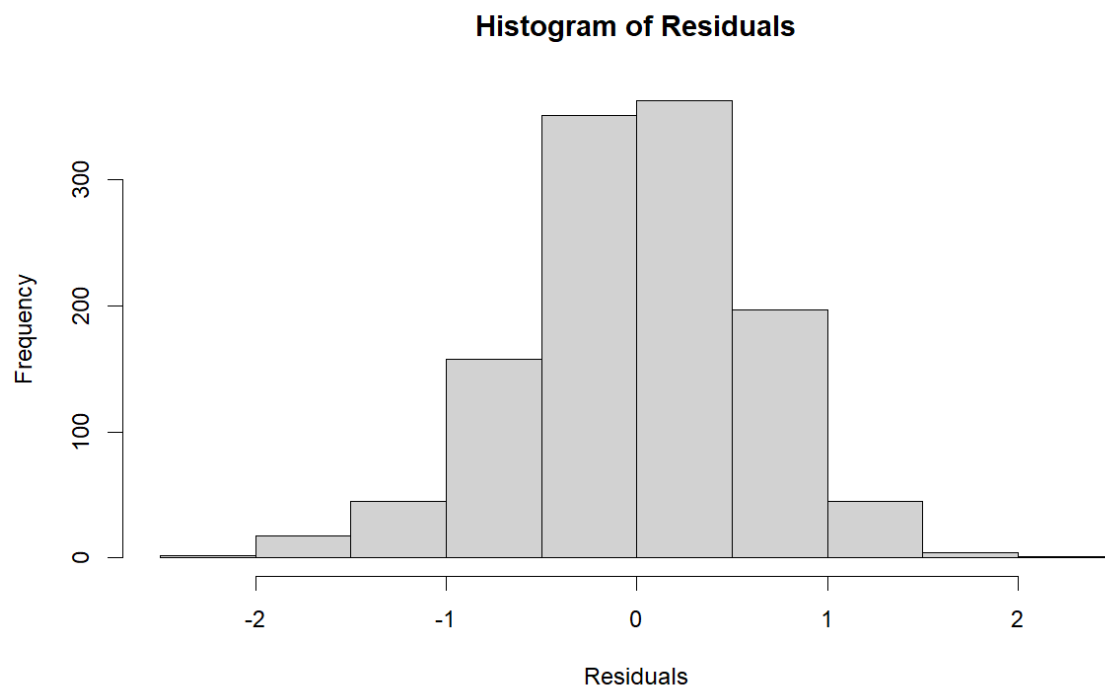
Or as Professor Munier says: “*Bref en attendant le bonheur, soyons heureux !*”

Appendices:

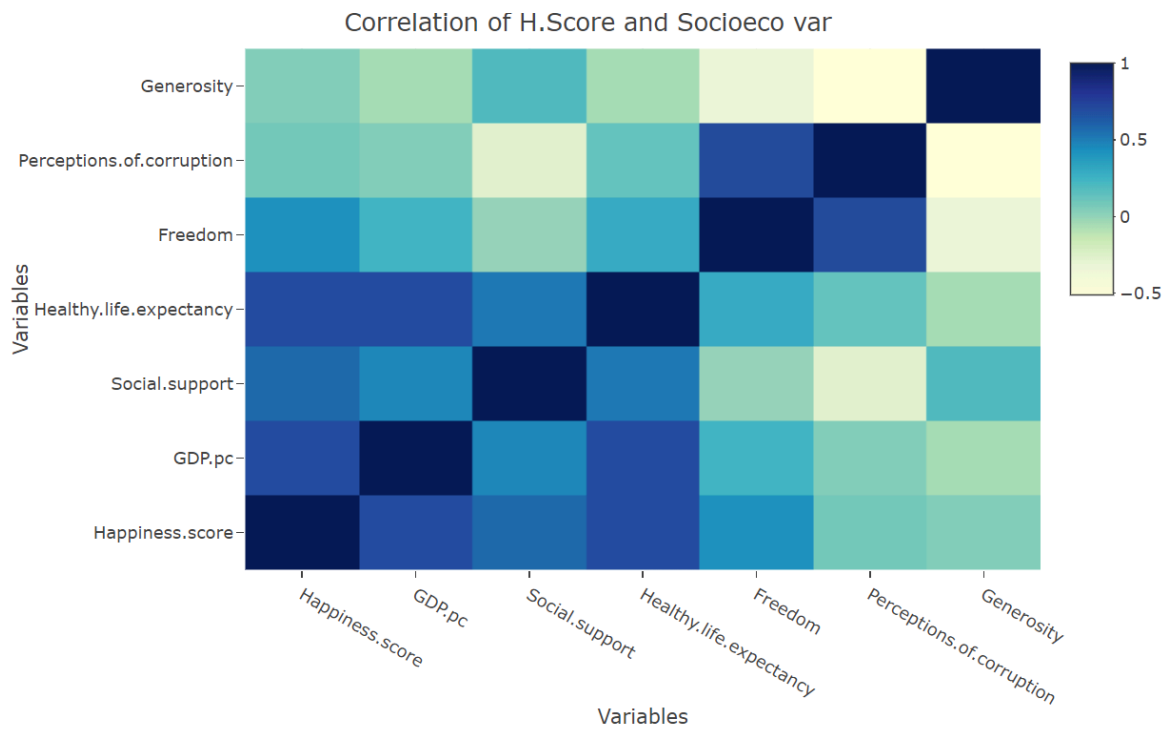
Appendix 1: Residuals Distribution for Linear Regression Model.



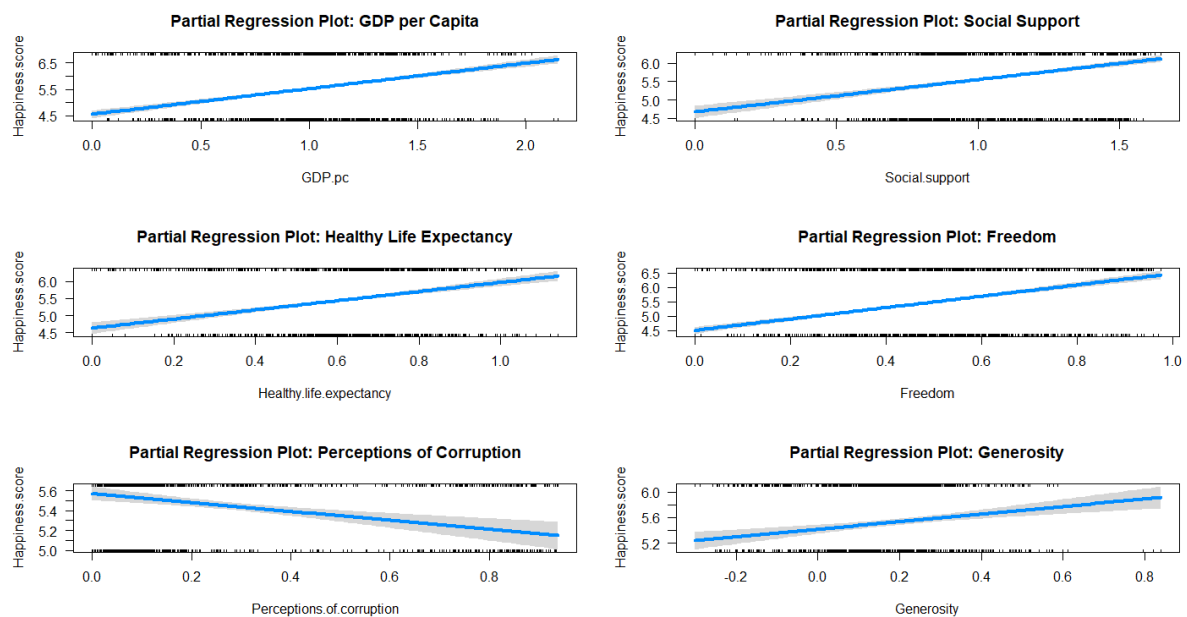
Appendix 2: Normality of Residuals for Linear Regression Model.



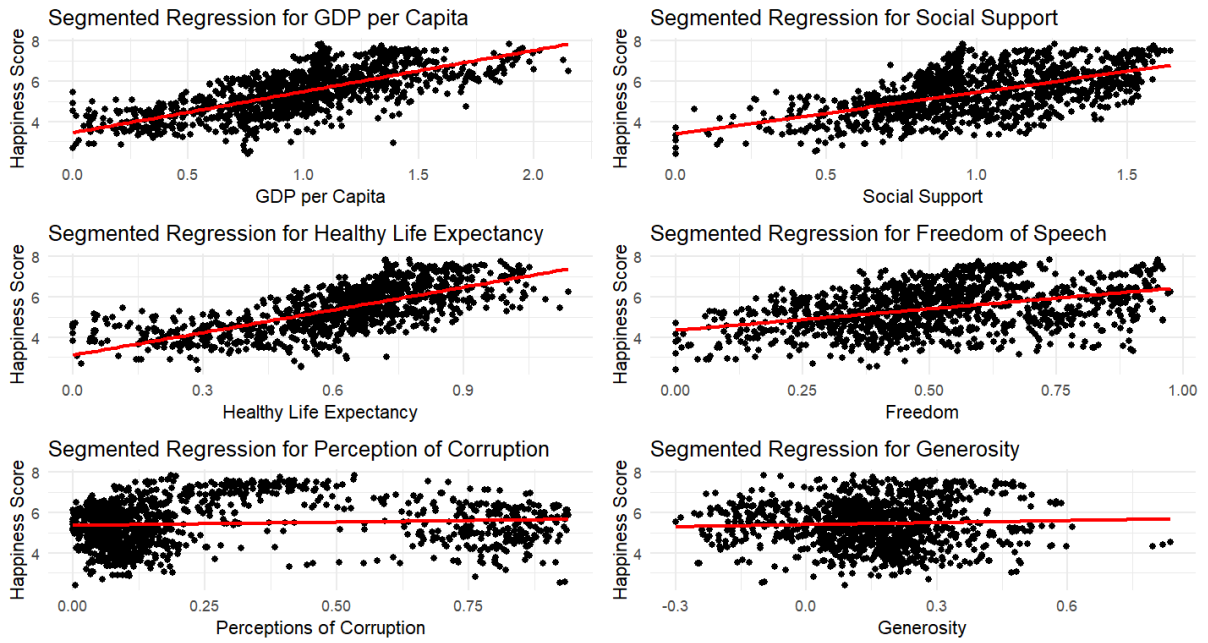
Appendix 3: Correlation Heatmap - Socioeconomic Factors and Happiness Scores.



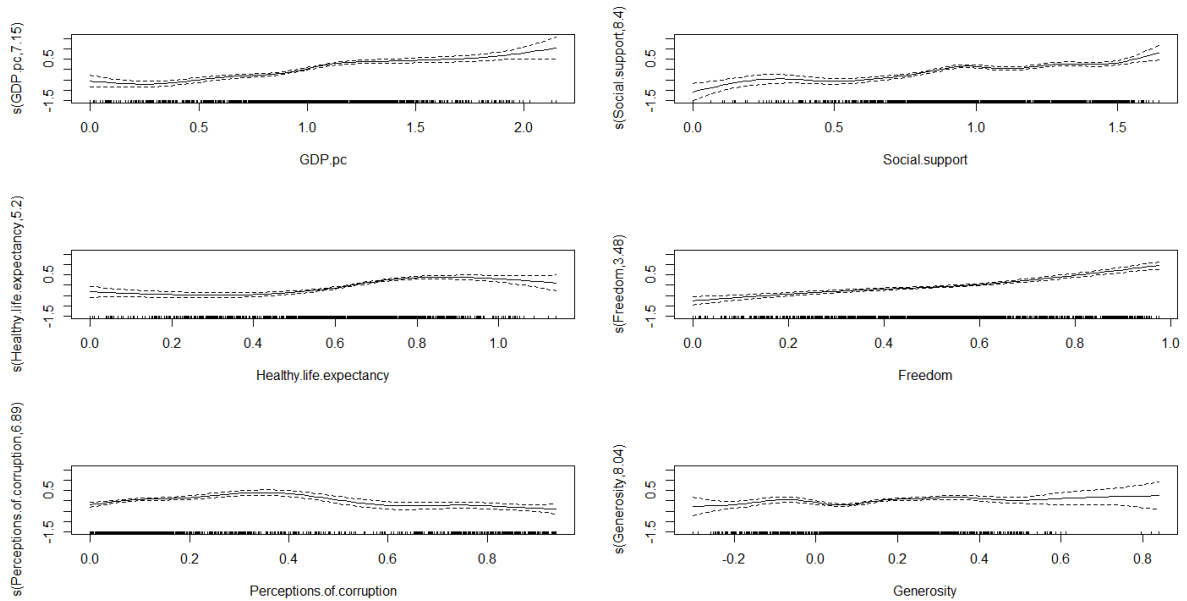
Appendix 4: Partial Regression (Linearity Questioning).



Appendix 5: Segmented Regression for Each Factor (BreakPoint Search).



Appendix 6: GAM regression.



Appendix 7-8: Rstudio Markdowns Attached to this report.

References:

“Multiple Measures of Family and Social Support as Predictors of Psychological Well-Being: An Additive Approach”, Hema O. Mason, 2016

“Psychosocial Factors Associated with Happiness”, Chloe R. Grabanski, Adedoyin Okanlawon, and Raina V. Lamade, 2022

“Economic growth and subjective well-being: reassessing the easterlin paradox”, Betsey Stevenson, Justin Wolfers, 2008

Socioeconomic Disparities in Health: Pathways and Policies”, Nancy E. Adler and Katherine Newman, 2002

“L'économie du bonheur: quelques faits saillants et données empiriques en Europe” Laetitia Dillenseger, Francis Munier, 2019

¹ *“Unemployment and (un)happiness: Life satisfaction approach to enhance policy efficiency for developing countries”, Khatai Aliyev, 2021*

² *“A longitudinal experimental study comparing the effectiveness of happiness-enhancing strategies in Anglo Americans and Asian Americans”, Julia K. Boehm, Sonja Lyubomirsky, Kennon M. Sheldon, 2010*

“Would you be happier living in a greener urban area? A fixed-effects analysis of panel data” Mathew P White , Ian Alcock, Benedict W Wheeler, Michael H Depledge, 2013

“Can and Should Happiness Be a Policy Goal?” Shigehiro Oishi, Ed Diener, 2014